



Off Grid Solar Battery Charger

Josh Benner

Abstract

This project involves the design and development of a 12 V solar-to-battery power system that efficiently converts solar energy into a regulated DC output suitable for off-grid applications. The system architecture integrates a solar panel, a buck converter, and a battery management system controlled by an embedded microcontroller running a Maximum PowerPoint Tracking algorithm. The initial phase of the project focuses on system modeling and simulation using MATLAB Simulink to evaluate converter performance, MPPT efficiency, and battery charging behavior under varying solar conditions. Simulation results demonstrate effective voltage regulation and energy transfer from the solar panel to the battery. These findings guide the hardware design and component selection for the next phase, which will involve constructing and testing a physical prototype. The project demonstrates how simulation driven design can accelerate the development of reliable, efficient solar power systems that combine power electronics and embedded control for renewable energy applications.

Background

The growing adoption of renewable energy has increased demand for compact photovoltaic (PV) systems capable of delivering reliable power in off-grid and portable applications. However, challenges such as changing irradiance, nonlinear PV characteristics, and conversion losses reduce overall system efficiency.

This project develops a solar-to-battery 12-V power system integrating PV generation, Maximum Power Point Tracking (MPPT), DC--DC conversion, and embedded control. MATLAB Simulink is used to model PV behavior, evaluate MPPT algorithms, and validate system performance before hardware implementation.

Prior research shows that MPPT methods such as Perturb & Observe (P&O), Incremental Conductance, and adaptive techniques improve power extraction under changing environmental conditions. Studies on converter topologies and battery management provide guidance for selecting efficient charging architectures and improving battery longevity.

Simulation studies demonstrate that MATLAB Simulink is an effective tool for PV system development and optimization. Building on this research, the project combines simulation-driven design with practical hardware implementation to create an efficient, low-cost, and portable renewable energy solution.

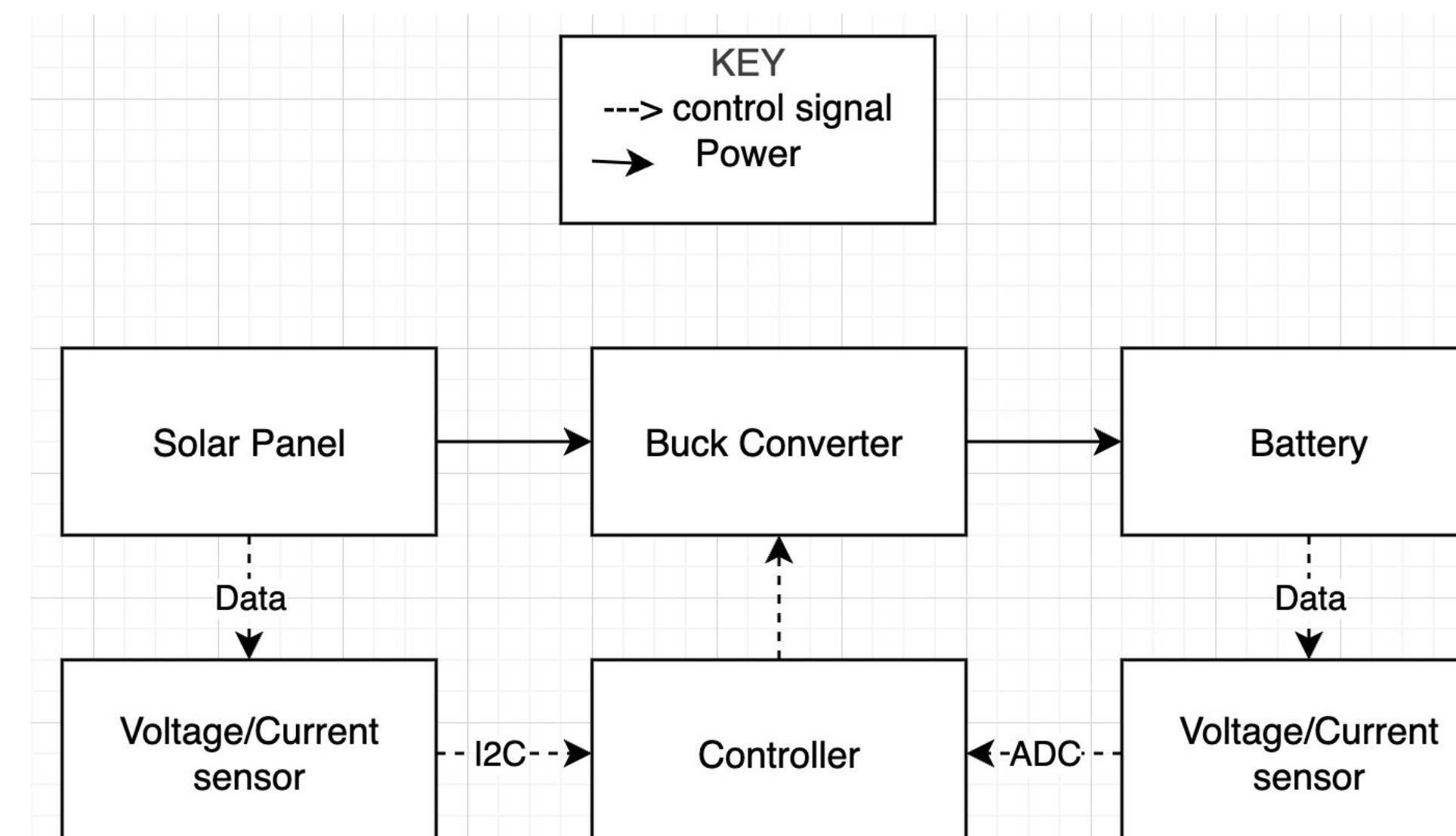
Methods and Materials

Materials

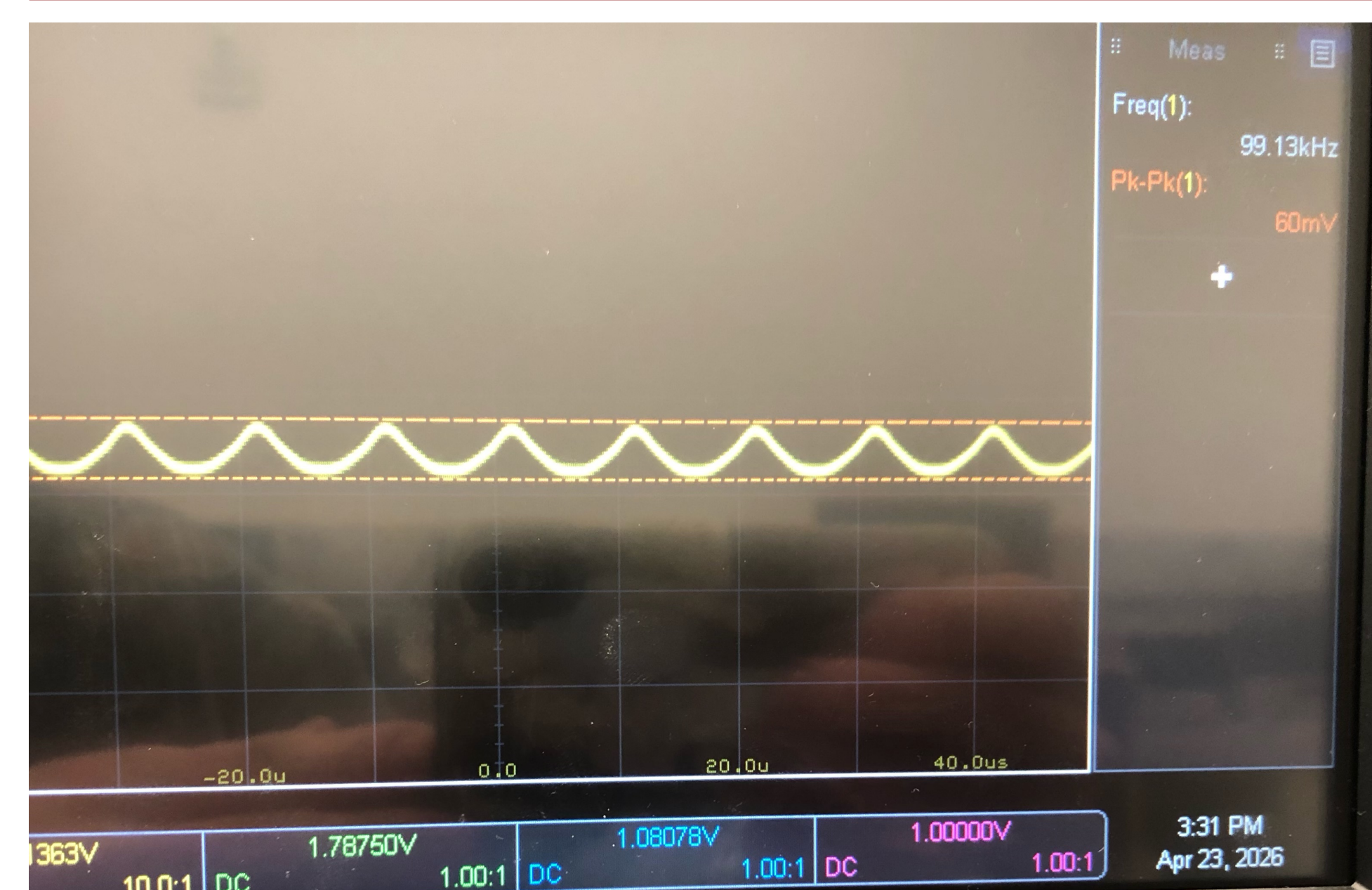
- Photovoltaic (PV) solar panel
- Sealed Lead Acid (SLA) battery
- DC-DC buck converter components
- Microcontroller and sensing circuitry
- MATLAB Simulink

Methods

1. Model PV panel behavior in MATLAB Simulink
2. Implement and compare MPPT algorithms
3. Simulate DC-DC converter performance
4. Develop battery charging control logic
5. Validate operation under varying irradiance conditions
6. Prepare for hardware implementation/testing

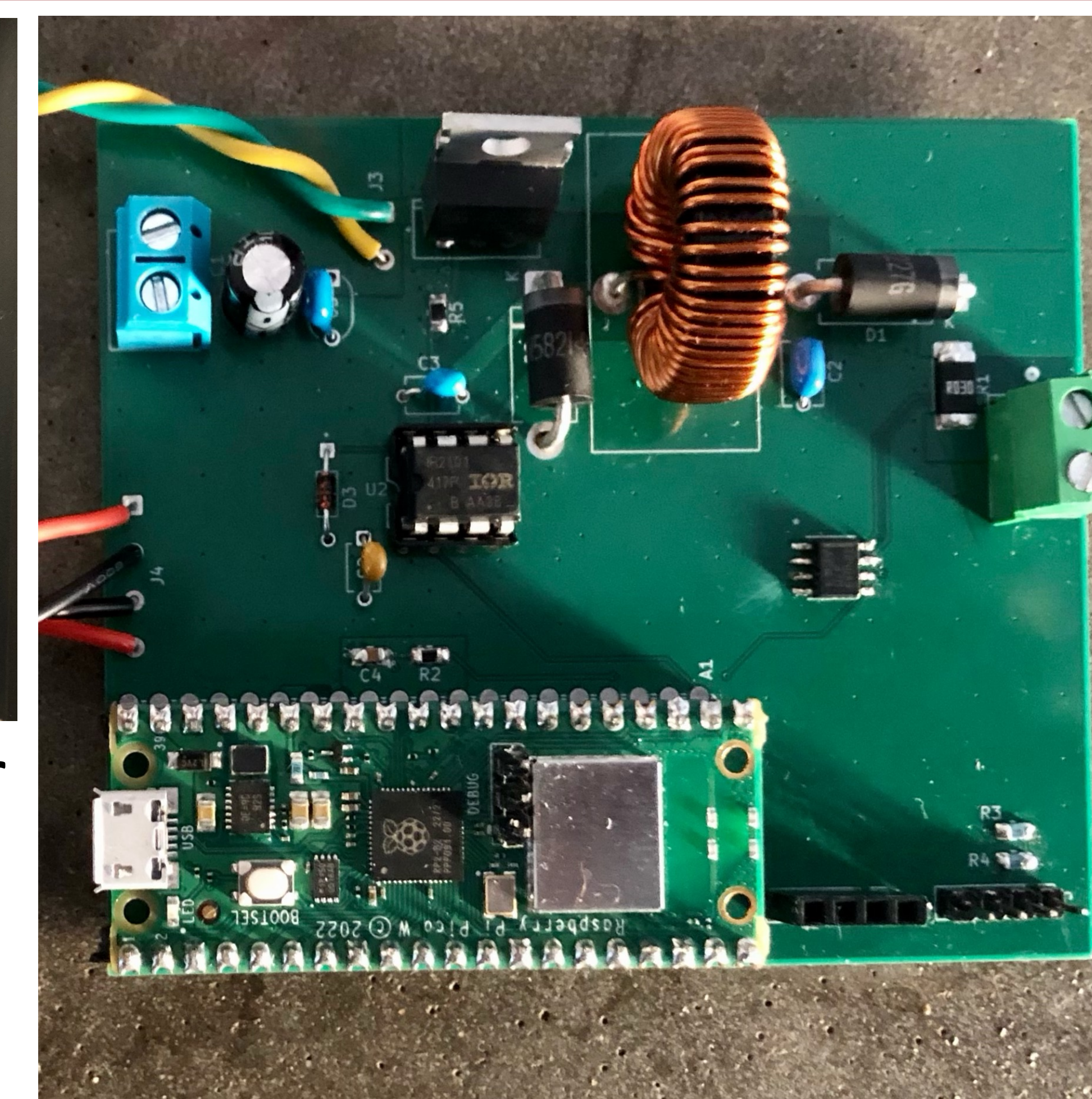


Results



Output Ripple of Buck Converter

- 90% efficiency
- Isolated noise



PCB

Future Direction

- Replace Raspberry Pi Pico with IC
- Replace Jumper wires with clips/Headers
- Re-design PCB and add silk screen for readability.

Acknowledgements

- Appreciation to the Electrical and Computer Engineering Department for providing laboratory resources and technical support.